SC306 Homework 4

Othar Zaldastani II

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Question 1

Creation of birth weight table

```
bw = matrix(c(110, 1073, 289, 15896), nrow=2, byrow = TRUE)
colnames(bw) = c('Low', 'Normal')
rownames(bw) = c('Cocaine', 'No Cocaine')
knitr::kable(bw)
```

	Low	Normal
Cocaine	110	1073
No Cocaine	289	15896

A) The incidence measure we can calculate given the data is incidence proportion. $IP = \frac{\text{Total number of new cases during interval}}{\text{Total population at risk}}$.

B) $IP_{Cocaine} = \frac{110}{110+289} = 0.276$

C) $IP_{Non-Cocaine} = \frac{1073}{1073+15896} = 0.0063$

D) The relative risk of low birth weight is $\frac{0.276}{0.063} = 4.38$.

E) A relative risk of 4.38 means mothers who used Cocaine during pregnancy had 4.38 times the risk of having a baby with low birth weight compared to mothers who did not use Cocaine during pregnancy.

F) Birth weight table

```
riskratio(bw, rev = "b")
```

```
## $data
##            Normal Low Total
## No Cocaine  15896 289 16185
## Cocaine      1073 110  1183
## Total       16969 399 17368
##
## $measure
```

```
##                          NA
## risk ratio with 95% C.I. estimate   lower    upper
##               No Cocaine 1.000000      NA       NA
##               Cocaine    5.207422 4.21475 6.433893
##
## $p.value
##             NA
## two-sided    midp.exact fisher.exact   chi.square
##   No Cocaine         NA           NA           NA
##   Cocaine             0 1.500218e-38 3.033658e-62
##
## $correction
## [1] FALSE
##
## attr(,"method")
## [1] "Unconditional MLE & normal approximation (Wald) CI"
```

The risk ratio is 5.21 with a 95% CI of [4.21 : 6.43]. The interpretation for the 95% confidence interval is that given our analysis of the data, we are 95% confident that the true risk ratio is between 4.21 and 6.43. Cocaine use is significantly associated with low birth weight, as shown by the value lower bound of our confidence interval considerably exceeding 1.

Question 2

```r
IPR.conf = function(a, b, c, d, conf) {
  ipr.hat = (a/(a+b)) / (c/(c+d))                  # Predicted IPR
  alpha = 1 - conf
  z = qnorm(1-alpha/2)

  var.ln.ipr = (b/(a*(a+b))) + (d/(c*(c+d)))       # Natural log of IPR variance
  se = sqrt(var.ln.ipr)                            # Standard error

  ln.lower = log(ipr.hat) - z * sqrt(var.ln.ipr)  # Upper/Lower Log Bounds
  ln.upper = log(ipr.hat) + z * sqrt(var.ln.ipr)

  lower = exp(ln.lower)                            # Upper/Lower Bounds
  upper = exp(ln.upper)

  data.frame(
    IPR = ipr.hat,
    "Lower Bound" = lower,
    "Upper Bound" = upper,
    "Confidence Interval" = conf,
    check.names = F
  )}
kable(IPR.conf(110, 1073, 289, 15896, .95), digits = 3)
```

IPR	Lower Bound	Upper Bound	Confidence Interval
5.207	4.215	6.434	0.95

The new interval is effectively identical to the interval from Problem 1.